_Experiment No. 3

Title

Deleting, Copying, Moving and Renaming Files in Linux

Aim

To understand and practice file management in Linux by using commands to delete, copy, move, and rename files and directories.

Theory

File Management Commands

Linux offers several commands to manage files:

- touch – Create empty files
- cp – Copy files or directories
- mv – Move or rename files
- rm – Delete files or directories
- cat – Display file contents

Input/Output Redirection

Redirection allows command outputs to be redirected to files instead of the terminal:

- > – Redirect output (overwrite)
- >> – Redirect output (append)
- < – Redirect input

These features are essential for scripting and automation.

Commands Used

touch	Create empty file
cp	Copy files
mv	Move or rename files
rm	Remove files

cat	Display file contents
>	Output redirection
>>	Append Output

Procedure / Steps

• Log in to the Linux system and open the terminal.

• Create a working directory:

```
┌──(setika㉿kali)-[~]
└─$ mkdir file_ops

┌──(setika㉿kali)-[~]
└─$ cd file_ops
```

• Create files using touch:

```
┌──(setika㉿kali)-[~/file_ops]
└─$ touch file1.txt file2.txt
```

• Add content to files using redirection:

```
┌──(setika㉿kali)-[~/file_ops]
└─$ echo "this is file1" > file1.txt

┌──(setika㉿kali)-[~/file_ops]
└─$ echo "this is file2" > file2.txt
```

• Display file contents:

```
┌──(setika㉿kali)-[~/file_ops]
└─$ cat file1.txt
this is file1
```

• Copy a file:

```
┌──(setika㉿kali)-[~/file_ops]
└─$ cp file1.txt copy_file1.txt
```

• Rename a file:

```
┌──(setika㉿kali)-[~/file_ops]
└─$ mv file2.txt renamed_file2.txt
```

• Move a file to another directory:

```
┌──(setika㉿kali)-[~/file_ops]
└─$ mkdir backup
mkdir: cannot create directory ‘backup’: File exists

┌──(setika㉿kali)-[~/file_ops]
└─$ mv copy_file1.txt backup/
```

• Append content to a file:

```
┌──(setika㉿kali)-[~/file_ops]
└─$ echo "additional content" >> file1.txt
```

• Delete a file:

```
┌──(setika㉿kali)-[~/file_ops]
└─$ rm renamed_file2.txt
```

• Verify changes using:

```
┌──(setika㉿kali)-[~/file_ops]
└─$ backup file1.txt
```

Outcomes

- Files created, copied, renamed, and deleted successfully.
- File contents displayed correctly.
- Output redirection works as expected.

Result

Linux file operations including copying, moving, renaming, deleting files, and output redirection were successfully performed using command-line utilities.